

## TURBOT BANK BUOY 62303, United Kingdom

WMO: 996860

Lat: 51.600N

Lon: 5.100W

Elev: 0

StdP: 101.33

Time Zone: 0.00 (EUW)

Period: 01-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 1.9        | 3.0 | -3.8                          | 2.7 | 3.8  | -2.4 | 3.1 | 4.4  | 16.4                  | 8.7  | 15.5 | 8.2  | 6.7                   | 60   | 0.764 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 1.7                    | 18.2            | 16.5 | 17.6 | 16.1 | 17.2 | 15.8 | 17.1                | 17.7 | 16.6 | 17.2 | 16.2 | 16.9 | 6.1                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.8                            | 12.0 | 17.4 | 16.2 | 11.6 | 17.0 | 15.8 | 11.2 | 16.7 | 48.0          | 17.7 | 46.6 | 17.3 | 45.4 | 17.0 | 20.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 15.4          | 14.1 | 12.9 | DB                         | 0.7  | 19.5               | 1.2 | 1.7                                                | -0.1 | 20.8       | -0.8 | 21.8       | -1.5 | 22.8       | -2.3 | 24.0 | (4) |
| (5)               |      |      | WB                         | -0.7 | 17.7               | 1.2 | 1.3                                                | -1.6 | 18.6       | -2.3 | 19.4       | -3.0 | 20.1       | -3.9 | 21.1 | (5) |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 11.4   | 8.1   | 7.4   | 8.0   | 9.4   | 11.1  | 13.4  | 15.2  | 15.9  | 15.2  | 13.3  | 11.2  | 8.8  |
|                                                                               | DBStd       | 3.39   | 2.33  | 2.02  | 1.48  | 1.31  | 1.24  | 1.09  | 1.06  | 0.94  | 1.14  | 1.75  | 2.05  | 2.49 |
|                                                                               | HDD10.0     | 292    | 63    | 73    | 63    | 26    | 3     | 0     | 0     | 0     | 0     | 2     | 12    | 50   |
|                                                                               | HDD18.3     | 2518   | 317   | 306   | 321   | 269   | 224   | 149   | 97    | 76    | 95    | 156   | 215   | 294  |
|                                                                               | CDD10.0     | 815    | 4     | 1     | 1     | 7     | 38    | 101   | 161   | 183   | 155   | 104   | 47    | 13   |
|                                                                               | CDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDH23.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 6.9    | 8.0   | 7.0   | 6.6   | 5.6   | 6.1   | 5.4   | 5.7   | 6.0   | 6.9   | 8.0   | 8.5   | 8.6  |
| Precipitation                                                                 | PrecAvg     | 1057   | 111   | 81    | 68    | 67    | 65    | 67    | 72    | 89    | 76    | 117   | 133   | 110  |
|                                                                               | PrecMax     | 1294   | 181   | 190   | 140   | 144   | 134   | 145   | 193   | 203   | 137   | 192   | 259   | 196  |
|                                                                               | PrecMin     | 891    | 11    | 19    | 22    | 23    | 7     | 11    | 16    | 25    | 9     | 33    | 73    | 28   |
|                                                                               | PrecStd     | 123    | 45    | 40    | 33    | 32    | 32    | 35    | 38    | 45    | 34    | 44    | 51    | 43   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 12.2  | 11.5  | 11.5  | 13.8  | 15.9  | 16.8  | 19.0  | 19.2  | 18.9  | 17.0  | 14.8  | 13.6 |
|                                                                               |             | MCWB   | 11.1  | 10.3  | 9.1   | 10.5  | 12.9  | 14.6  | 16.4  | 17.3  | 16.6  | 15.9  | 12.7  | 12.5 |
|                                                                               | 2%          | DB     | 11.7  | 10.8  | 10.7  | 12.4  | 14.2  | 15.8  | 17.8  | 18.3  | 17.5  | 16.3  | 14.4  | 12.7 |
|                                                                               |             | MCWB   | 10.6  | 9.8   | 9.0   | 10.3  | 12.1  | 14.3  | 16.1  | 16.7  | 16.1  | 15.2  | 12.5  | 11.5 |
|                                                                               | 5%          | DB     | 11.2  | 10.3  | 10.2  | 11.7  | 13.2  | 15.2  | 17.2  | 17.8  | 17.1  | 15.9  | 14.1  | 12.2 |
|                                                                               |             | MCWB   | 10.1  | 9.3   | 8.8   | 10.1  | 11.8  | 13.9  | 15.9  | 16.4  | 15.7  | 14.8  | 12.3  | 10.9 |
|                                                                               | 10%         | DB     | 10.8  | 9.9   | 9.8   | 11.1  | 12.7  | 14.8  | 16.7  | 17.4  | 16.7  | 15.5  | 13.7  | 11.9 |
|                                                                               |             | MCWB   | 9.6   | 8.9   | 8.5   | 9.7   | 11.4  | 13.6  | 15.5  | 16.0  | 15.3  | 14.3  | 11.9  | 10.5 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 11.6  | 10.6  | 10.1  | 11.8  | 13.2  | 15.4  | 17.2  | 17.8  | 17.3  | 16.4  | 14.1  | 13.0 |
|                                                                               |             | MCDB   | 11.8  | 11.1  | 10.4  | 12.4  | 14.4  | 16.2  | 17.7  | 18.5  | 18.1  | 16.6  | 14.5  | 13.5 |
|                                                                               | 2%          | WB     | 11.0  | 10.1  | 9.6   | 11.0  | 12.6  | 14.7  | 16.6  | 17.3  | 16.7  | 15.7  | 13.5  | 12.0 |
|                                                                               |             | MCDB   | 11.5  | 10.5  | 10.1  | 11.9  | 13.6  | 15.4  | 17.3  | 17.9  | 17.2  | 16.1  | 14.2  | 12.6 |
|                                                                               | 5%          | WB     | 10.5  | 9.6   | 9.2   | 10.4  | 12.2  | 14.4  | 16.2  | 16.8  | 16.2  | 15.2  | 12.9  | 11.3 |
|                                                                               |             | MCDB   | 11.1  | 10.2  | 9.8   | 11.2  | 13.0  | 15.0  | 17.0  | 17.5  | 16.8  | 15.7  | 13.9  | 12.1 |
|                                                                               | 10%         | WB     | 9.9   | 9.1   | 8.9   | 10.0  | 11.7  | 14.0  | 15.7  | 16.4  | 15.7  | 14.6  | 12.4  | 10.8 |
|                                                                               |             | MCDB   | 10.7  | 9.8   | 9.7   | 10.7  | 12.5  | 14.6  | 16.6  | 17.1  | 16.4  | 15.4  | 13.4  | 11.7 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 2.3   | 2.2   | 2.0   | 2.1   | 1.9   | 1.8   | 1.7   | 1.7   | 1.9   | 2.1   | 2.2   | 2.6  |
|                                                                               |             | MCDBR  | 2.4   | 2.3   | 2.1   | 3.1   | 2.8   | 2.5   | 2.3   | 2.2   | 2.2   | 2.0   | 2.0   | 2.9  |
|                                                                               |             | MCWBR  | 3.0   | 2.4   | 2.2   | 2.6   | 2.4   | 2.4   | 2.1   | 2.1   | 2.3   | 2.6   | 2.8   | 3.7  |
|                                                                               | 5% WB       | MCDBR  | 2.3   | 2.1   | 1.9   | 2.3   | 2.4   | 2.2   | 2.1   | 1.9   | 1.9   | 1.9   | 2.0   | 2.7  |
| MCWBR                                                                         |             | 3.1    | 2.5   | 2.2   | 2.2   | 2.3   | 2.3   | 2.4   | 2.1   | 2.2   | 2.4   | 2.7   | 3.0   | 3.7  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.338  | 0.349 | 0.377 | 0.388 | 0.398 | 0.383 | 0.389 | 0.372 | 0.367 | 0.359 | 0.336 | 0.328 |      |
|                                                                               | taud        | 2.329  | 2.377 | 2.327 | 2.332 | 2.353 | 2.426 | 2.414 | 2.477 | 2.476 | 2.444 | 2.384 | 2.306 |      |
|                                                                               | Ebn at Noon | 656    | 767   | 812   | 849   | 857   | 872   | 860   | 858   | 820   | 747   | 657   | 597   |      |
|                                                                               | Edh at Noon | 65     | 81    | 102   | 114   | 117   | 109   | 109   | 97    | 88    | 75    | 61    | 57    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 0.74   | 1.53  | 2.77  | 4.42  | 5.42  | 5.89  | 5.38  | 4.49  | 3.33  | 1.84  | 0.93  | 0.56  |      |
|                                                                               | RadStd      | 0.07   | 0.14  | 0.24  | 0.31  | 0.34  | 0.50  | 0.44  | 0.30  | 0.25  | 0.15  | 0.07  | 0.07  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (42 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page